

# Derivation of the Spectral Photon Yield for a Table-Top X-Ray Source

**Fourier convention and transcription note.** Throughout, the Fourier transform is defined by

$$\mathbf{F}(\mathbf{r}, \omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{F}(\mathbf{r}, t) e^{-i\omega t} dt, \quad \mathbf{F}(\mathbf{r}, t) = \int_{-\infty}^{\infty} \mathbf{F}(\mathbf{r}, \omega) e^{i\omega t} d\omega. \quad (1)$$

## 1 Charge conservation and electromagnetic energy balance

The total charge contained in a fixed volume  $V$  is

$$Q(t) = \int_V \rho(\mathbf{r}, t) d\tau. \quad (2)$$

The rate at which charge leaves the volume is the outward current flux,

$$\frac{dQ}{dt} = - \oint_{\partial V} \mathbf{J} \cdot d\mathbf{a}. \quad (3)$$

Using the divergence theorem,

$$\int_V \frac{\partial \rho}{\partial t} d\tau = - \int_V \nabla \cdot \mathbf{J} d\tau, \quad (4)$$

so the local continuity equation is

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.} \quad (5)$$

The electric and magnetic field energies are

$$W_E = \frac{\varepsilon_0}{2} \int_V |\mathbf{E}|^2 d\tau, \quad W_M = \frac{1}{2\mu_0} \int_V |\mathbf{B}|^2 d\tau. \quad (6)$$

Hence the electromagnetic energy density is

$$u = \frac{dW}{d\tau} = \frac{1}{2} \left( \varepsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0} |\mathbf{B}|^2 \right). \quad (7)$$

For a charge  $q$  moving with velocity  $\mathbf{v}$ , the infinitesimal work done by the Lorentz force is

$$dW = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} dt \quad (8)$$

$$= q\mathbf{E} \cdot \mathbf{v} dt = \mathbf{F} \cdot d\boldsymbol{\ell}, \quad (9)$$

because  $(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} = 0$ : the magnetic force does no work. For a small volume element,  $q = \rho d\tau$  and  $\mathbf{J} = \rho\mathbf{v}$ , giving

$$\left. \frac{dW}{dt} \right|_{d\tau} = \mathbf{E} \cdot \mathbf{J} d\tau. \quad (10)$$

Therefore, over a finite volume,

$$\frac{dW}{dt} = \int_V \mathbf{E} \cdot \mathbf{J} d\tau. \quad (11)$$

From the Ampère–Maxwell law,

$$\mathbf{J} = \frac{1}{\mu_0} \nabla \times \mathbf{B} - \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \quad (12)$$

so

$$\mathbf{E} \cdot \mathbf{J} = \frac{1}{\mu_0} \mathbf{E} \cdot (\nabla \times \mathbf{B}) - \varepsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t}. \quad (13)$$

Use the vector identity

$$\nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{B}), \quad (14)$$

together with Faraday’s law,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (15)$$

It follows that

$$\mathbf{E} \cdot (\nabla \times \mathbf{B}) = -\mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{B}). \quad (16)$$

Also,

$$\mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} |\mathbf{E}|^2, \quad \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} |\mathbf{B}|^2. \quad (17)$$

Substitution into Eq. (11) gives

$$\frac{dW}{dt} = -\frac{d}{dt} \int_V \frac{1}{2} \left( \varepsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0} |\mathbf{B}|^2 \right) d\tau - \frac{1}{\mu_0} \int_V \nabla \cdot (\mathbf{E} \times \mathbf{B}) d\tau \quad (18)$$

$$= -\frac{d}{dt} \int_V u d\tau - \oint_{\partial V} \mathbf{S} \cdot d\mathbf{a}, \quad (19)$$

where

$$\boxed{\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}} \quad (20)$$

is the Poynting vector. The local form of the energy-balance equation is

$$\boxed{\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J}.} \quad (21)$$

In a charge-free region,  $\mathbf{J} = 0$ , so

$$\frac{\partial u}{\partial t} = -\nabla \cdot \mathbf{S}. \quad (22)$$

For a far-field plane wave propagating along  $\hat{\mathbf{n}}$ ,

$$\mathbf{E} \perp \hat{\mathbf{n}}, \quad \mathbf{B} = \frac{1}{c} \hat{\mathbf{n}} \times \mathbf{E}. \quad (23)$$

Therefore,

$$\mathbf{S} \cdot \hat{\mathbf{n}} = \frac{1}{\mu_0 c} |\mathbf{E}|^2 = \varepsilon_0 c |\mathbf{E}(\mathbf{r}, t)|^2. \quad (24)$$

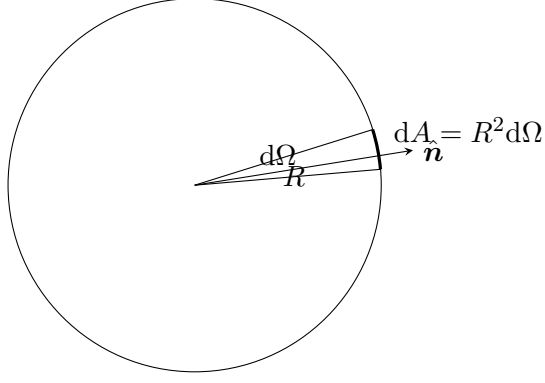


Figure 1: Far-field surface element subtending a solid angle  $d\Omega$ .

## 2 Radiated energy and spectral photon yield

At a distant observation sphere of radius  $R$ , a solid-angle element  $d\Omega$  corresponds to

$$dA = R^2 d\Omega. \quad (25)$$

The energy radiated through  $dA$  during  $dt$  is

$$dW = (\mathbf{S} \cdot \hat{\mathbf{n}}) dA dt. \quad (26)$$

Integrating over all time and using Eq. (24),

$$\frac{dW}{d\Omega} = \varepsilon_0 c R^2 \int_{-\infty}^{\infty} |\mathbf{E}(\mathbf{r}, t)|^2 dt. \quad (27)$$

Parseval's identity for the convention in Eq. (1) gives

$$\int_{-\infty}^{\infty} |\mathbf{E}(\mathbf{r}, t)|^2 dt = 2\pi \int_{-\infty}^{\infty} |\mathbf{E}(\mathbf{r}, \omega)|^2 d\omega \quad (28)$$

$$= 4\pi \int_0^{\infty} |\mathbf{E}(\mathbf{r}, \omega)|^2 d\omega, \quad (29)$$

where the second equality uses the reality condition  $\mathbf{E}(\mathbf{r}, -\omega) = \mathbf{E}(\mathbf{r}, \omega)^*$ . Thus,

$$\frac{dW}{d\Omega} = 4\pi \varepsilon_0 c R^2 \int_0^{\infty} |\mathbf{E}(\mathbf{r}, \omega)|^2 d\omega, \quad (30)$$

and the spectral radiated energy is

$$\boxed{\frac{d^2 W}{d\Omega d\omega} = 4\pi \varepsilon_0 c R^2 |\mathbf{E}(\mathbf{r}, \omega)|^2.} \quad (31)$$

A photon of angular frequency  $\omega$  has energy  $\hbar\omega$ , so

$$dW = \hbar\omega dN. \quad (32)$$

Combining this with Eq. (31),

$$\boxed{\frac{d^2 N}{d\Omega d\omega} = \frac{4\pi \varepsilon_0 c R^2}{\hbar\omega} |\mathbf{E}(\mathbf{r}, \omega)|^2.} \quad (33)$$

### 3 Electric field in terms of a dyadic Green tensor

Let

$$\mathbf{D}(\mathbf{r}, \omega) = \varepsilon_0 \varepsilon_r(\mathbf{r}, \omega) \mathbf{E}(\mathbf{r}, \omega), \quad \mathbf{B} = \mu_0 \mathbf{H}. \quad (34)$$

Using the Fourier convention in Eq. (1), Maxwell's curl equations become

$$\nabla \times \mathbf{E}(\mathbf{r}, \omega) = -i\omega \mathbf{B}(\mathbf{r}, \omega), \quad (35)$$

$$\nabla \times \mathbf{H}(\mathbf{r}, \omega) = \mathbf{J}(\mathbf{r}, \omega) + i\omega \mathbf{D}(\mathbf{r}, \omega). \quad (36)$$

Taking the curl of Faraday's law,

$$\nabla \times \nabla \times \mathbf{E} = -i\omega \mu_0 \nabla \times \mathbf{H} \quad (37)$$

$$= -i\omega \mu_0 (\mathbf{J} + i\omega \mathbf{D}) \quad (38)$$

$$= -i\omega \mu_0 \mathbf{J} + \omega^2 \mu_0 \varepsilon_0 \varepsilon_r \mathbf{E}. \quad (39)$$

Since  $c^{-2} = \mu_0 \varepsilon_0$ ,

$$\left[ \nabla \times \nabla \times - \frac{\omega^2}{c^2} \varepsilon_r(\mathbf{r}, \omega) \right] \mathbf{E}(\mathbf{r}, \omega) = -i\omega \mu_0 \mathbf{J}(\mathbf{r}, \omega). \quad (40)$$

Define the wave operator

$$\mathcal{L} = \nabla \times \nabla \times - \frac{\omega^2}{c^2} \varepsilon_r(\mathbf{r}, \omega), \quad (41)$$

and its dyadic Green tensor  $\mathbf{G}$  by

$$\mathcal{L}_{\mathbf{r}} \mathbf{G}(\mathbf{r}, \mathbf{r}', \omega) = \delta^{(3)}(\mathbf{r} - \mathbf{r}') \mathbb{I}. \quad (42)$$

Applying the inverse operator to Eq. (40) gives

$$\mathbf{E}(\mathbf{r}, \omega) = -i\omega \mu_0 \int \mathbf{G}(\mathbf{r}, \mathbf{r}', \omega) \cdot \mathbf{J}(\mathbf{r}', \omega) d^3 r'. \quad (43)$$

Indeed, applying  $\mathcal{L}_{\mathbf{r}}$  to the right-hand side and using Eq. (42) reproduces Eq. (40).

### 4 Vacuum Helmholtz Green function and its far-field form

Let

$$k = \frac{\omega}{c}, \quad g_0(\mathbf{r}, \mathbf{r}') = \frac{e^{-ik|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}. \quad (44)$$

This outgoing scalar Helmholtz Green function obeys

$$(\nabla^2 + k^2)g_0(\mathbf{r}, \mathbf{r}') = -\delta^{(3)}(\mathbf{r} - \mathbf{r}'). \quad (45)$$

The corresponding vacuum dyadic Green tensor is

$$\mathbf{G}_0 = \left( \mathbb{I} + \frac{\nabla \nabla}{k^2} \right) g_0, \quad (\nabla \nabla g_0)_{ij} = \partial_i \partial_j g_0. \quad (46)$$

Let the observation point be

$$\mathbf{r} = R\hat{\mathbf{n}}, \quad R \gg |\mathbf{r}'|. \quad (47)$$

Then

$$|\mathbf{r} - \mathbf{r}'| = |R\hat{\mathbf{n}} - \mathbf{r}'| \simeq R - \hat{\mathbf{n}} \cdot \mathbf{r}', \quad (48)$$

so

$$e^{-ik|\mathbf{r}-\mathbf{r}'|} \simeq e^{-ikR} e^{ik\hat{\mathbf{n}} \cdot \mathbf{r}'}, \quad \frac{1}{|\mathbf{r} - \mathbf{r}'|} \simeq \frac{1}{R}. \quad (49)$$

Consequently,

$$g_0(R\hat{\mathbf{n}}, \mathbf{r}') \simeq \frac{e^{-ikR}}{4\pi R} e^{ik\hat{\mathbf{n}} \cdot \mathbf{r}'}. \quad (50)$$

To leading order in  $1/R$ ,

$$\nabla g_0 \simeq -ik\hat{\mathbf{n}} g_0, \quad \nabla \nabla g_0 \simeq -k^2(\hat{\mathbf{n}} \otimes \hat{\mathbf{n}})g_0. \quad (51)$$

Substituting into Eq. (46),

$$\boxed{\mathbf{G}_0(R\hat{\mathbf{n}}, \mathbf{r}') \simeq \frac{e^{-ikR}}{4\pi R} (\mathbb{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}) e^{ik\hat{\mathbf{n}} \cdot \mathbf{r}'}.} \quad (52)$$

The operator

$$\mathbb{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}} \quad (53)$$

is the transverse projector because, for any vector  $\mathbf{a}$ ,

$$(\mathbb{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}) \mathbf{a} = \mathbf{a} - \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{a}). \quad (54)$$

Let  $\hat{\mathbf{e}}_1$  and  $\hat{\mathbf{e}}_2$  be two mutually orthogonal polarization vectors satisfying

$$\hat{\mathbf{e}}_s \cdot \hat{\mathbf{n}} = 0, \quad s = 1, 2. \quad (55)$$

Then

$$\mathbb{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}} = \sum_{s=1}^2 \hat{\mathbf{e}}_s \otimes \hat{\mathbf{e}}_s. \quad (56)$$

Writing  $\mathbf{k} = k\hat{\mathbf{n}}$ , Eq. (52) becomes

$$\boxed{\mathbf{G}_0(R\hat{\mathbf{n}}, \mathbf{r}') \simeq \frac{e^{-ikR}}{4\pi R} \sum_{s=1}^2 \hat{\mathbf{e}}_s \otimes [\hat{\mathbf{e}}_s e^{ik\hat{\mathbf{n}} \cdot \mathbf{r}'}].} \quad (57)$$

## 5 Lorentz reciprocity and the reciprocal field

For two monochromatic field systems  $(\mathbf{E}_1, \mathbf{H}_1, \mathbf{J}_1)$  and  $(\mathbf{E}_2, \mathbf{H}_2, \mathbf{J}_2)$  at the same frequency in a reciprocal anisotropic material, Lorentz reciprocity gives

$$\oint_{\partial V} (\mathbf{E}_1 \times \mathbf{H}_2 - \mathbf{E}_2 \times \mathbf{H}_1) \cdot d\mathbf{a} = \int_V (\mathbf{E}_2 \cdot \mathbf{J}_1 - \mathbf{E}_1 \cdot \mathbf{J}_2) d^3r. \quad (58)$$

Choose system 1 to be the electron-emitted field,

$$(\mathbf{E}_1, \mathbf{H}_1, \mathbf{J}_1) = (\mathbf{E}_{\text{em}}, \mathbf{H}_{\text{em}}, \mathbf{J}), \quad (59)$$

and system 2 to be a source-free reciprocal field associated with a far detector channel of propagation direction  $\mathbf{k} = k\hat{\mathbf{n}}$  and polarization  $s$ ,

$$(\mathbf{E}_2, \mathbf{H}_2, \mathbf{J}_2) = (\mathbf{E}_{\mathbf{k}s}^{\text{rec}}, \mathbf{H}_{\mathbf{k}s}^{\text{rec}}, \mathbf{0}). \quad (60)$$

Equation (58) then reduces to

$$\oint_{\partial V} (\mathbf{E}_{\text{em}} \times \mathbf{H}_{\mathbf{k}s}^{\text{rec}} - \mathbf{E}_{\mathbf{k}s}^{\text{rec}} \times \mathbf{H}_{\text{em}}) \cdot d\mathbf{a} = \int_V \mathbf{E}_{\mathbf{k}s}^{\text{rec}} \cdot \mathbf{J} d^3r. \quad (61)$$

The surface term extracts the field emitted into the specified far-field detector channel.

Motivated by the vacuum result in Eq. (57), the far-field Green tensor in the material is written as

$$\boxed{\mathbf{G}(R\hat{\mathbf{n}}, \mathbf{r}', \omega) \simeq \frac{e^{-ikR}}{4\pi R} \sum_s \hat{\mathbf{e}}_s \otimes \mathbf{E}_{\mathbf{k}s}^{\text{rec}}(\mathbf{r}', \omega).} \quad (62)$$

For a periodic lattice, the reciprocal field is represented in the handwritten notes as

$$\mathbf{E}_{\mathbf{k}s}^{\text{rec}}(\mathbf{r}', \omega) = \hat{\mathbf{e}}_s e^{i\mathbf{k} \cdot \mathbf{r}'} + \sum_{\mathbf{g} \neq \mathbf{0}} \mathbf{E}_{\mathbf{g}s} e^{i(\mathbf{k} + \mathbf{g}) \cdot \mathbf{r}'}, \quad (63)$$

where the first term is the incident plane wave and the remaining terms describe lattice-scattered components. This expansion is taken as given in the handwritten derivation.

## 6 Electron current and channel-resolved radiation amplitude

For an electron of charge  $-e$  following the trajectory  $\mathbf{r}_e(t)$ , with velocity  $\mathbf{v}(t) = \dot{\mathbf{r}}_e(t)$ , the current density is

$$\mathbf{J}(\mathbf{r}, t) = -e\mathbf{v}(t) \delta^{(3)}(\mathbf{r} - \mathbf{r}_e(t)). \quad (64)$$

Its Fourier transform is

$$\mathbf{J}(\mathbf{r}, \omega) = -\frac{e}{2\pi} \int_{-\infty}^{\infty} \mathbf{v}(t) \delta^{(3)}(\mathbf{r} - \mathbf{r}_e(t)) e^{-i\omega t} dt. \quad (65)$$

Define the trajectory integral for polarization channel  $s$  by

$$\mathcal{I}_s(\omega) = \int_{-\infty}^{\infty} \mathbf{E}_{\mathbf{k}s}^{\text{rec}}(\mathbf{r}_e(t), \omega) \cdot \mathbf{v}(t) e^{-i\omega t} dt. \quad (66)$$

Then

$$\int \mathbf{E}_{\mathbf{k}s}^{\text{rec}}(\mathbf{r}, \omega) \cdot \mathbf{J}(\mathbf{r}, \omega) d^3r = -\frac{e}{2\pi} \mathcal{I}_s(\omega). \quad (67)$$

Substituting Eq. (62) into Eq. (43),

$$\mathbf{E}(R\hat{\mathbf{n}}, \omega) \simeq -\frac{i\omega\mu_0 e^{-ikR}}{4\pi R} \sum_s \hat{\mathbf{e}}_s \int \mathbf{E}_{\mathbf{k}s}^{\text{rec}}(\mathbf{r}', \omega) \cdot \mathbf{J}(\mathbf{r}', \omega) d^3r' \quad (68)$$

$$= \frac{i\omega\mu_0 e^{-ikR}}{8\pi^2 R} \sum_s \hat{\mathbf{e}}_s \mathcal{I}_s(\omega). \quad (69)$$

Therefore, the component with polarization  $s$  is

$$\mathbf{E}_s(R\hat{\mathbf{n}}, \omega) = \frac{i\omega\mu_0 e^{-ikR}}{8\pi^2 R} \hat{\mathbf{e}}_s \mathcal{I}_s(\omega), \quad (70)$$

and

$$R^2 |\mathbf{E}_s(R\hat{\mathbf{n}}, \omega)|^2 = \frac{e^2 \omega^2 \mu_0^2}{64\pi^4} |\mathcal{I}_s(\omega)|^2. \quad (71)$$

Combining Eqs. (33) and (71),

$$\frac{d^2 N}{d\Omega d\omega} = \frac{4\pi\epsilon_0 c}{\hbar\omega} \frac{e^2 \omega^2 \mu_0^2}{64\pi^4} |\mathcal{I}_s|^2 \quad (72)$$

$$= \frac{e^2 \omega \mu_0^2 \epsilon_0 c}{16\pi^3 \hbar} |\mathcal{I}_s|^2. \quad (73)$$

Using

$$\mu_0 \epsilon_0 = \frac{1}{c^2}, \quad \mu_0^2 \epsilon_0 c = \frac{1}{\epsilon_0 c^3}, \quad (74)$$

Eq. (73) becomes

$$\frac{d^2 N}{d\Omega d\omega} = \frac{e^2 \omega}{16\pi^3 \epsilon_0 \hbar c^3} |\mathcal{I}_s|^2. \quad (75)$$

Introducing the fine-structure constant

$$\alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c}, \quad (76)$$

we obtain

$$\boxed{\frac{d^2 N}{d\Omega d\omega} = \frac{\alpha}{4\pi^2} \frac{\omega}{c^2} |\mathcal{I}_s(\omega)|^2.} \quad (77)$$

Finally, substituting Eq. (66),

$$\boxed{\frac{d^2 N}{d\Omega d\omega} = \frac{\alpha}{4\pi^2} \frac{\omega}{c^2} \left| \int_{-\infty}^{\infty} \mathbf{v}(t) \cdot \mathbf{E}_{\mathbf{k}s}^{\text{rec}}(\mathbf{r}_e(t), \omega) e^{-i\omega t} dt \right|^2.} \quad (78)$$

## 7 Segmented trajectories and averaging over electrons

For the  $i$ -th segment of an electron path, parameterized over

$$0 \leq t \leq t_L, \quad (79)$$

the contribution written in the handwritten derivation is

$$\left[ \frac{d^2 N}{d\Omega d\omega} \right]_i = \frac{\alpha}{4\pi^2} \frac{\omega}{c^2} \left| \int_0^{t_L} \mathbf{v}_i(t) \cdot \mathbf{E}_{\mathbf{k}s}^{\text{rec}}(\mathbf{r}_i(t), \omega) e^{-i\omega t} dt \right|^2. \quad (80)$$

For the  $j$ -th electron, summing over its trajectory segments gives

$$\left[ \frac{d^2 N}{d\Omega d\omega} \right]_j = \frac{\alpha}{4\pi^2} \frac{\omega}{c^2} \sum_{i=1}^{N_{\text{coll}}^{(j)}} \left| \int_0^{t_L} \mathbf{v}_i^{(j)}(t) \cdot \mathbf{E}_{\mathbf{k}s}^{\text{rec}}(\mathbf{r}_i^{(j)}(t), \omega) e^{-i\omega t} dt \right|^2. \quad (81)$$

Averaging over  $N_e$  electrons yields

$$\boxed{\frac{d^2 N}{d\Omega d\omega} = \frac{1}{N_e} \frac{\alpha}{4\pi^2} \frac{\omega}{c^2} \sum_{j=1}^{N_e} \sum_{i=1}^{N_{\text{coll}}^{(j)}} \left| \int_0^{t_L} \mathbf{v}_i^{(j)}(t) \cdot \mathbf{E}_{\mathbf{k}s}^{\text{rec}}(\mathbf{r}_i^{(j)}(t), \omega) e^{-i\omega t} dt \right|^2.} \quad (82)$$

## 8 Fourier expansion of the susceptibility

The susceptibility function is periodic under a Bravais-lattice translation  $\mathbf{R}$ :

$$\chi(\mathbf{r} + \mathbf{R}, \omega) = \chi(\mathbf{r}, \omega). \quad (83)$$

Therefore it can be expanded in reciprocal-lattice vectors  $\mathbf{g}$  as

$$\chi(\mathbf{r}, \omega) = \sum_{\mathbf{g}} \chi_{\mathbf{g}}(\omega) e^{i\mathbf{g} \cdot \mathbf{r}}. \quad (84)$$

Multiply by  $e^{-i\mathbf{g}' \cdot \mathbf{r}}$  and integrate over one primitive cell of volume  $V$ :

$$\frac{1}{V} \int_V \chi(\mathbf{r}, \omega) e^{-i\mathbf{g}' \cdot \mathbf{r}} d^3r = \sum_{\mathbf{g}} \chi_{\mathbf{g}}(\omega) \frac{1}{V} \int_V e^{i(\mathbf{g} - \mathbf{g}') \cdot \mathbf{r}} d^3r \quad (85)$$

$$= \sum_{\mathbf{g}} \chi_{\mathbf{g}}(\omega) \delta_{\mathbf{g}\mathbf{g}'}. \quad (86)$$

Hence

$$\boxed{\chi_{\mathbf{g}}(\omega) = \frac{1}{V} \int_V \chi(\mathbf{r}, \omega) e^{-i\mathbf{g} \cdot \mathbf{r}} d^3r}. \quad (87)$$

We now derive  $\chi_{\mathbf{g}}$  from the motion of a bound electron and the corresponding bound charge density in the lattice.

## 9 Bound-electron response

Near a stable atomic equilibrium, expand the binding potential about  $\mathbf{r} = 0$ :

$$U(\mathbf{r}) = U(0) + \nabla U|_0 \cdot \mathbf{r} + \frac{1}{2} \sum_{ij} \left. \frac{\partial^2 U}{\partial r_i \partial r_j} \right|_0 r_i r_j + \dots \quad (88)$$

At equilibrium the force vanishes, so

$$\nabla U|_0 = 0. \quad (89)$$

Define the Hessian

$$K_{ij} \equiv \left. \frac{\partial^2 U}{\partial r_i \partial r_j} \right|_0. \quad (90)$$

For an isotropic binding potential,  $\mathbf{K} = k\mathbf{I}$ , and therefore

$$U(\mathbf{r}) \simeq U(0) + \frac{1}{2} k |\mathbf{r}|^2. \quad (91)$$

The restoring force is

$$\mathbf{F} = -\nabla U = -k\mathbf{r} = m_e \ddot{\mathbf{r}}, \quad (92)$$

so that

$$k = m_e \omega_0^2, \quad (93)$$

where  $\omega_0$  is the natural resonance frequency of the bound electron. Thus

$$U(\mathbf{r}) \simeq U(0) + \frac{1}{2} m_e \omega_0^2 |\mathbf{r}|^2, \quad \mathbf{F}_U(\mathbf{r}) = -m_e \omega_0^2 \mathbf{r}. \quad (94)$$

Now apply an external electric field  $\mathbf{E}$ . The electric force on the electron is

$$\mathbf{F}_e = -e\mathbf{E}. \quad (95)$$

Hence (before damping is included),

$$m_e \ddot{\mathbf{x}} = -m_e \omega_0^2 \mathbf{x} - e\mathbf{E}. \quad (96)$$

For a monochromatic field, use the convention

$$\mathbf{E}(t) = \text{Re}[\mathbf{E}_\omega e^{-i\omega t}], \quad \mathbf{x}(t) = \text{Re}[\mathbf{x}_\omega e^{-i\omega t}]. \quad (97)$$

To select a unique steady-state long-time response, include a damping rate  $\Gamma$ :

$$m_e (\ddot{\mathbf{x}} + \Gamma \dot{\mathbf{x}} + \omega_0^2 \mathbf{x}) = -e\mathbf{E}(t). \quad (98)$$

For the underdamped case, the homogeneous motion has the form

$$\mathbf{x}_h(t) = e^{-\Gamma t/2} [\mathbf{A} \cos(\Omega t) + \mathbf{B} \sin(\Omega t)], \quad \Omega = \sqrt{\omega_0^2 - \frac{\Gamma^2}{4}}, \quad (99)$$

so that  $\mathbf{x}_h(t) \rightarrow 0$  as  $t \rightarrow \infty$ .

Keeping only the driven-frequency component gives

$$m_e (-\omega^2 - i\Gamma\omega + \omega_0^2) \mathbf{x}_\omega = -e\mathbf{E}_\omega, \quad (100)$$

therefore

$$\mathbf{x}_\omega = -\frac{e}{m_e (-\omega^2 - i\Gamma\omega + \omega_0^2)} \mathbf{E}_\omega. \quad (101)$$

The dipole moment of the displaced electron is

$$\mathbf{p}_\omega = -e\mathbf{x}_\omega = \frac{e^2}{m_e (-\omega^2 - i\Gamma\omega + \omega_0^2)} \mathbf{E}_\omega. \quad (102)$$

In the high-frequency (Thomson-response) limit,

$$\omega^2 \gg \omega_0^2, \quad \omega \gg \Gamma, \quad (103)$$

we obtain

$$\boxed{\mathbf{p}_\omega \simeq -\frac{e^2}{m_e \omega^2} \mathbf{E}_\omega}, \quad \mathbf{x}_\omega \simeq \frac{e}{m_e \omega^2} \mathbf{E}_\omega. \quad (104)$$

Let  $n_e(\mathbf{r})$  be the equilibrium electron number density, so that

$$\rho(\mathbf{r}) = -e n_e(\mathbf{r}). \quad (105)$$

If the electron cloud is displaced slightly by  $\mathbf{x}(t)$ , then

$$\frac{\rho'(\mathbf{r})}{-e} = n_e(\mathbf{r} - \mathbf{x}) \quad (106)$$

$$\simeq n_e(\mathbf{r}) - \mathbf{x} \cdot \nabla n_e(\mathbf{r}). \quad (107)$$

Therefore

$$\delta\rho = \rho' - \rho = e \mathbf{x} \cdot \nabla n_e(\mathbf{r}) \equiv \rho_b = -\nabla \cdot \mathbf{P}. \quad (108)$$

Since  $\mathbf{x}$  is not a function of the spatial coordinate  $\mathbf{r}$ ,

$$\nabla \cdot \mathbf{x} = 0, \quad (109)$$

and hence

$$-\nabla \cdot \mathbf{P} = e \mathbf{x} \cdot \nabla n_e(\mathbf{r}) + e n_e(\mathbf{r})(\nabla \cdot \mathbf{x}) \quad (110)$$

$$= \nabla \cdot [e n_e(\mathbf{r}) \mathbf{x}]. \quad (111)$$

Thus

$$\boxed{\mathbf{P} = -e n_e(\mathbf{r}) \mathbf{x}}. \quad (112)$$

Using the high-frequency displacement,

$$\mathbf{P} = -\frac{e^2 n_e(\mathbf{r})}{m_e \omega^2} \mathbf{E}(\mathbf{r}, \omega). \quad (113)$$

For a linear dielectric,

$$\mathbf{P} = \varepsilon_0 \chi(\mathbf{r}, \omega) \mathbf{E}, \quad (114)$$

so

$$\boxed{\chi(\mathbf{r}, \omega) = -\frac{e^2}{\varepsilon_0 m_e \omega^2} n_e(\mathbf{r})}. \quad (115)$$

It follows that

$$\chi_{\mathbf{g}}(\omega) = \frac{1}{V} \int_V \chi(\mathbf{r}, \omega) e^{-i\mathbf{g} \cdot \mathbf{r}} d^3r \quad (116)$$

$$= -\frac{e^2}{\varepsilon_0 m_e \omega^2} \left[ \frac{1}{V} \int_V n_e(\mathbf{r}) e^{-i\mathbf{g} \cdot \mathbf{r}} d^3r \right] \quad (117)$$

$$\equiv -\frac{e^2}{\varepsilon_0 m_e \omega^2} n_{\mathbf{g}}. \quad (118)$$

## 10 Electron-density Fourier coefficient and structure factor

Now we determine  $n_{\mathbf{g}}$ . Let the equilibrium position of atom  $a$  inside one unit cell be  $\boldsymbol{\tau}_a$ , and let  $n_a(\mathbf{u})$  denote the electron number density of atom  $a$ , normalized by

$$\int n_a(\mathbf{u}) d^3u = Z_a, \quad (119)$$

where  $Z_a$  is the electron number of the neutral atom. The crystal electron density is

$$n_e(\mathbf{r}) = \sum_{\mathbf{R}} \sum_a n_a(\mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_a). \quad (120)$$

For a crystal containing  $N_c$  unit cells, write the Fourier coefficient as an average over the total crystal volume  $N_c V$ :

$$n_{\mathbf{g}} = \frac{1}{N_c V} \int_{N_c V} n_e(\mathbf{r}) e^{-i\mathbf{g} \cdot \mathbf{r}} d^3r \quad (121)$$

$$= \frac{1}{N_c V} \sum_{\mathbf{R}, a} \int n_a(\mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_a) e^{-i\mathbf{g} \cdot \mathbf{r}} d^3r. \quad (122)$$

Let

$$\mathbf{u} = \mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_a, \quad \mathrm{d}^3u = \mathrm{d}^3r. \quad (123)$$

Then

$$n_{\mathbf{g}} = \frac{1}{N_c V} \sum_{\mathbf{R}, a} \int n_a(\mathbf{u}) e^{-i\mathbf{g} \cdot (\mathbf{u} + \mathbf{R} + \boldsymbol{\tau}_a)} \mathrm{d}^3u \quad (124)$$

$$= \frac{1}{N_c V} \sum_{\mathbf{R}, a} e^{-i\mathbf{g} \cdot \mathbf{R}} e^{-i\mathbf{g} \cdot \boldsymbol{\tau}_a} \int n_a(\mathbf{u}) e^{-i\mathbf{g} \cdot \mathbf{u}} \mathrm{d}^3u. \quad (125)$$

Define the atomic form factor

$$f_a(\mathbf{g}) \equiv \int n_a(\mathbf{u}) e^{-i\mathbf{g} \cdot \mathbf{u}} \mathrm{d}^3u. \quad (126)$$

Because  $\mathbf{g}$  is a reciprocal-lattice vector,

$$e^{-i\mathbf{g} \cdot \mathbf{R}} = 1, \quad \sum_{\mathbf{R}} e^{-i\mathbf{g} \cdot \mathbf{R}} = N_c. \quad (127)$$

Therefore

$$n_{\mathbf{g}} = \frac{1}{V} \sum_a f_a(\mathbf{g}) e^{-i\mathbf{g} \cdot \boldsymbol{\tau}_a}. \quad (128)$$

Define the structure factor

$$S(\mathbf{g}) \equiv \sum_a f_a(\mathbf{g}) e^{-i\mathbf{g} \cdot \boldsymbol{\tau}_a}, \quad (129)$$

so that

$$\boxed{n_{\mathbf{g}} = \frac{S(\mathbf{g})}{V}}. \quad (130)$$

If atom  $a$  undergoes an instantaneous thermal displacement  $\mathbf{u}_a$ , then the thermally averaged structure factor contains

$$\langle e^{-i\mathbf{g} \cdot \mathbf{u}_a} \rangle. \quad (131)$$

For harmonic Gaussian thermal motion this is conventionally written

$$\langle e^{-i\mathbf{g} \cdot \mathbf{u}_a} \rangle = e^{-W_a(\mathbf{g})}, \quad (132)$$

where  $W_a$  is the Debye–Waller exponent. Thus, in general,

$$\boxed{\chi_{\mathbf{g}}(\omega) = -\frac{e^2}{\varepsilon_0 m_e \omega^2 V} \sum_a f_a(\mathbf{g}) e^{-i\mathbf{g} \cdot \boldsymbol{\tau}_a} e^{-W_a(\mathbf{g})}}. \quad (133)$$

If all atoms share the same Debye–Waller factor,  $W_a(\mathbf{g}) = W(\mathbf{g})$ , then

$$\boxed{\chi_{\mathbf{g}}(\omega) = -\frac{e^2}{\varepsilon_0 m_e \omega^2} \frac{S(\mathbf{g})}{V} e^{-W(\mathbf{g})}}. \quad (4)$$

## 11 Source-free Maxwell equation

Now consider the source-free electromagnetic field (the step labelled “Eq. (3)” in the handwritten notes):

$$\rho_f = 0, \quad \mathbf{J}_f = 0. \quad (134)$$

With the  $e^{-i\omega t}$  convention,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = i\omega \mathbf{B}, \quad (135)$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} = -i\omega \mathbf{D}. \quad (136)$$

Since  $\mathbf{B} = \mu_0 \mathbf{H}$ ,

$$\nabla \times \mathbf{B} = -i\omega \mu_0 \mathbf{D}. \quad (137)$$

Therefore

$$\nabla \times \nabla \times \mathbf{E} = i\omega \nabla \times \mathbf{B} \quad (138)$$

$$= \omega^2 \mu_0 \mathbf{D} \quad (139)$$

$$= \omega^2 \mu_0 \varepsilon_0 [1 + \chi(\mathbf{r}, \omega)] \mathbf{E}. \quad (140)$$

Using

$$k_0 \equiv \frac{\omega}{c}, \quad c^{-2} = \mu_0 \varepsilon_0, \quad (141)$$

we obtain

$$\boxed{(\nabla \times \nabla \times -k_0^2) \mathbf{E} = k_0^2 \chi(\mathbf{r}, \omega) \mathbf{E}}. \quad (142)$$

We now solve this equation perturbatively. Introduce a bookkeeping parameter  $\eta$ :

$$\chi \rightarrow \eta \chi, \quad \mathbf{E} = \mathbf{E}_0 + \eta \mathbf{E}_1 + \eta^2 \mathbf{E}_2 + \cdots. \quad (143)$$

Then

$$(\nabla \times \nabla \times -k_0^2) (\mathbf{E}_0 + \eta \mathbf{E}_1 + \eta^2 \mathbf{E}_2 + \cdots) = \eta k_0^2 \chi (\mathbf{E}_0 + \eta \mathbf{E}_1 + \cdots). \quad (144)$$

At zeroth order,

$$(\nabla \times \nabla \times -k_0^2) \mathbf{E}_0 = 0. \quad (145)$$

A transverse plane-wave solution is

$$\mathbf{E}_0(\mathbf{r}) = \hat{\mathbf{e}}_s e^{i\mathbf{k} \cdot \mathbf{r}}, \quad |\mathbf{k}| = k_0 = \frac{\omega}{c}, \quad \mathbf{k} \cdot \hat{\mathbf{e}}_s = 0. \quad (146)$$

At first order,

$$(\nabla \times \nabla \times -k_0^2) \mathbf{E}_1 = k_0^2 \chi \mathbf{E}_0. \quad (147)$$

Using

$$\chi(\mathbf{r}, \omega) = \sum_{\mathbf{g}} \chi_{\mathbf{g}}(\omega) e^{i\mathbf{g} \cdot \mathbf{r}}, \quad (148)$$

the source term becomes

$$k_0^2 \chi \mathbf{E}_0 = k_0^2 \sum_{\mathbf{g}} \chi_{\mathbf{g}}(\omega) \hat{\mathbf{e}}_s e^{i(\mathbf{k}+\mathbf{g}) \cdot \mathbf{r}}. \quad (149)$$

$$(\nabla \times \nabla \times -k_0^2) \mathbf{E}_1 = k_0^2 \sum_{\mathbf{g} \neq 0} \chi_{\mathbf{g}} \hat{\mathbf{e}}_s e^{i(\mathbf{k}+\mathbf{g}) \cdot \mathbf{r}}. \quad (150)$$

Take the ansatz

$$\mathbf{E}_1 = \sum_{\mathbf{g} \neq 0} \mathbf{C}_g e^{i(\mathbf{k}+\mathbf{g}) \cdot \mathbf{r}}. \quad (151)$$

For a given reciprocal vector define

$$\mathbf{q} \equiv \mathbf{k} + \mathbf{g}. \quad (152)$$

Then

$$\nabla \times (\mathbf{C}_g e^{i\mathbf{q} \cdot \mathbf{r}}) = i\mathbf{q} \times \mathbf{C}_g e^{i\mathbf{q} \cdot \mathbf{r}}, \quad (153)$$

$$\nabla \times \nabla \times (\mathbf{C}_g e^{i\mathbf{q} \cdot \mathbf{r}}) = -\mathbf{q} \times (\mathbf{q} \times \mathbf{C}_g) e^{i\mathbf{q} \cdot \mathbf{r}} \quad (154)$$

$$= [q^2 \mathbf{C}_g - \mathbf{q}(\mathbf{q} \cdot \mathbf{C}_g)] e^{i\mathbf{q} \cdot \mathbf{r}}. \quad (155)$$

Therefore the coefficient of each  $e^{i\mathbf{q} \cdot \mathbf{r}}$  satisfies

$$\boxed{(q^2 - k_0^2) \mathbf{C}_g - \mathbf{q}(\mathbf{q} \cdot \mathbf{C}_g) = k_0^2 \chi_g \hat{\mathbf{e}}_s}. \quad (156)$$

Dotting Eq. (156) with  $\mathbf{q}$ ,

$$(q^2 - k_0^2)(\mathbf{q} \cdot \mathbf{C}_g) - q^2(\mathbf{q} \cdot \mathbf{C}_g) = k_0^2 \chi_g \mathbf{q} \cdot \hat{\mathbf{e}}_s, \quad (157)$$

$$-k_0^2 \mathbf{q} \cdot \mathbf{C}_g = k_0^2 \chi_g \mathbf{q} \cdot \hat{\mathbf{e}}_s. \quad (158)$$

Hence

$$-\mathbf{q} \cdot \mathbf{C}_g = \chi_g \mathbf{q} \cdot \hat{\mathbf{e}}_s. \quad (159)$$

Since  $\mathbf{q} = \mathbf{k} + \mathbf{g}$  and the incident plane wave is transverse,

$$\mathbf{k} \cdot \hat{\mathbf{e}}_s = 0, \quad (160)$$

we have

$$\mathbf{q} \cdot \hat{\mathbf{e}}_s = \mathbf{g} \cdot \hat{\mathbf{e}}_s, \quad (161)$$

and therefore

$$-\mathbf{q} \cdot \mathbf{C}_g = \chi_g \mathbf{g} \cdot \hat{\mathbf{e}}_s. \quad (162)$$

Substituting this back into Eq. (156),

$$(q^2 - k_0^2) \mathbf{C}_g + \mathbf{q} \chi_g (\mathbf{g} \cdot \hat{\mathbf{e}}_s) = k_0^2 \chi_g \hat{\mathbf{e}}_s, \quad (163)$$

so

$$\boxed{\mathbf{C}_g = \chi_g \frac{k_0^2 \hat{\mathbf{e}}_s - \mathbf{q}(\mathbf{g} \cdot \hat{\mathbf{e}}_s)}{q^2 - k_0^2}}. \quad (164)$$

Equivalently, with  $\mathbf{q} = \mathbf{k} + \mathbf{g}$ ,  $k_0 = \omega/c$ , and

$$k_g^2 \equiv |\mathbf{k} + \mathbf{g}|^2, \quad (165)$$

this becomes

$$\boxed{\mathbf{C}_g = -\chi_g(\omega) \left[ \frac{(\mathbf{k} + \mathbf{g})(\mathbf{g} \cdot \hat{\mathbf{e}}_s) - \frac{\omega^2}{c^2} \hat{\mathbf{e}}_s}{k_g^2 - \frac{\omega^2}{c^2}} \right]}. \quad (166)$$

Finally,

$$\mathbf{E} = \mathbf{E}_0 + \mathbf{E}_1, \quad (167)$$

with

$$\mathbf{E} = \hat{\mathbf{e}}_s e^{i\mathbf{k}\cdot\mathbf{r}} + \sum_{\mathbf{g} \neq 0} C_{\mathbf{g}} e^{i(\mathbf{k}+\mathbf{g})\cdot\mathbf{r}}, \quad (168)$$

and therefore

$$\mathbf{E}(\mathbf{r}) = \hat{\mathbf{e}}_s e^{i\mathbf{k}\cdot\mathbf{r}} - \sum_{\mathbf{g} \neq 0} \chi_{\mathbf{g}}(\omega) \left[ \frac{(\mathbf{k} + \mathbf{g})(\mathbf{g} \cdot \hat{\mathbf{e}}_s) - \frac{\omega^2}{c^2} \hat{\mathbf{e}}_s}{k_{\mathbf{g}}^2 - \frac{\omega^2}{c^2}} \right] e^{i(\mathbf{k}+\mathbf{g})\cdot\mathbf{r}}. \quad (169)$$